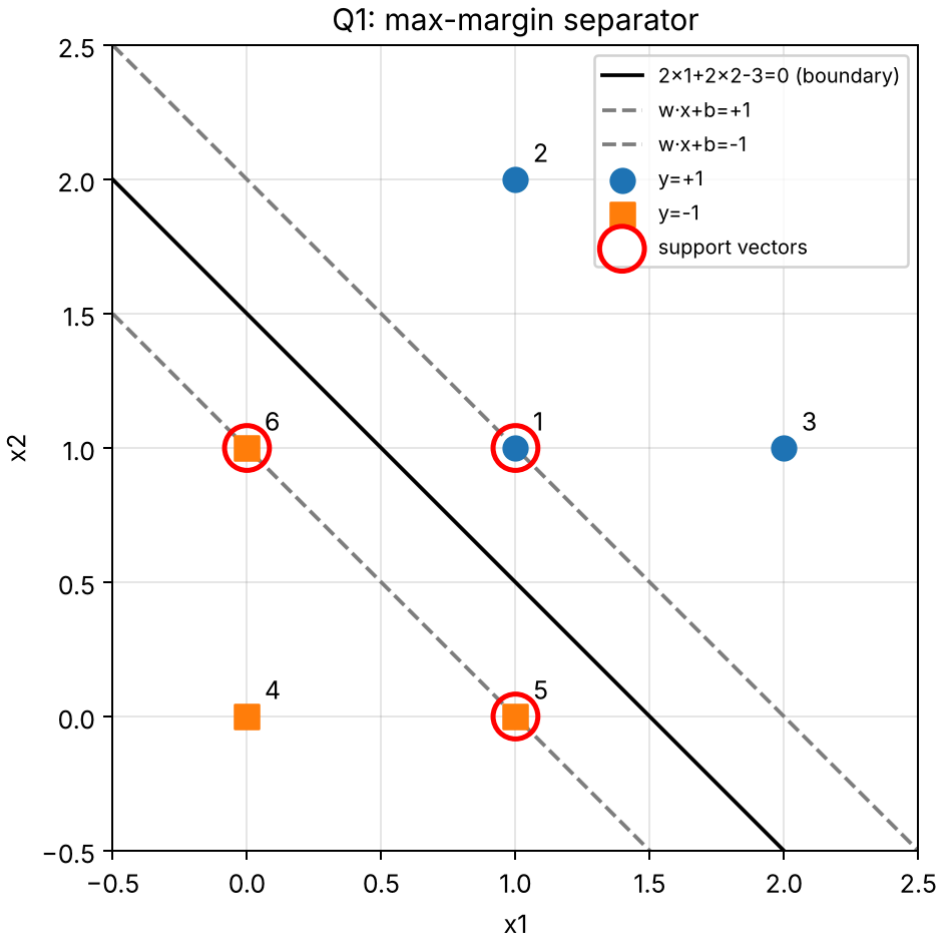
CSE 242 — Assignment 3

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Q1: Hard-margin SVM

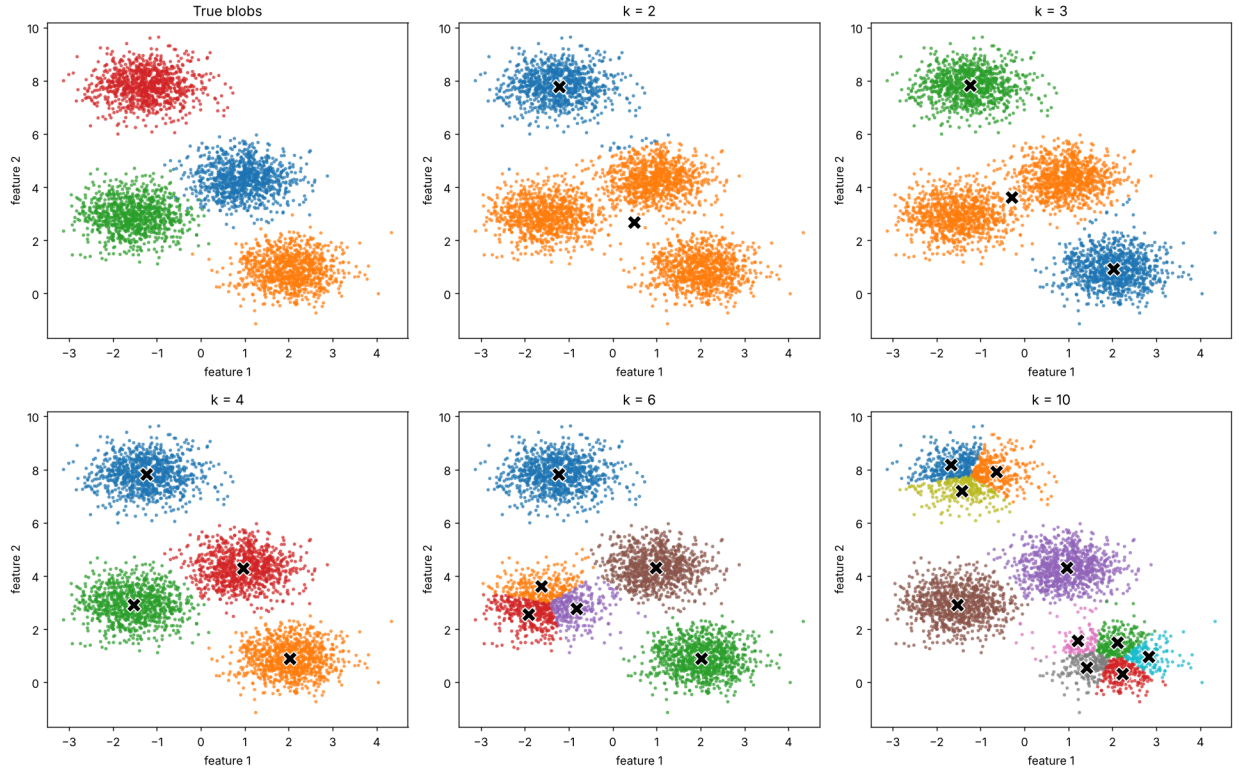
(a) Support vectors: **points 1, 5, 6**. Negatives all have x1+x2 ≤ 1 and positives x1+x2 ≥ 2, so the widest gap is between those two lines. Point 1 (1,1) is the closest positive; points 5 (1,0) and 6 (0,1) are the closest negatives. Points 2, 3, 4 are farther away and do not affect the boundary.
(b) By symmetry w = (a, a). Setting y(w·x + b) = 1 on the support vectors: point 1 gives 2a + b = 1, and point 5 (or 6) gives a + b = −1. So a = 2, b = −3.
Boundary: **2x1 + 2x2 − 3 = 0** (w = (2, 2), b = −3). All other points give y(w·x+b) = 3 ≥ 1.
Geometric margin (boundary to closest points): **1/||w|| = 1/(2√2) = √2/4 ≈ 0.354**. Full width between the ±1 lines: 2/||w|| ≈ 0.707.



Q2: K-means

(a) Assignment step: Euclidean distance to each centroid, then argmin. Ties go to the lowest-index centroid. Update step: each centroid becomes the mean of its points; an empty cluster keeps its old centroid (this handles duplicate initial centroids).
(b) Seed 0, 100 iterations.

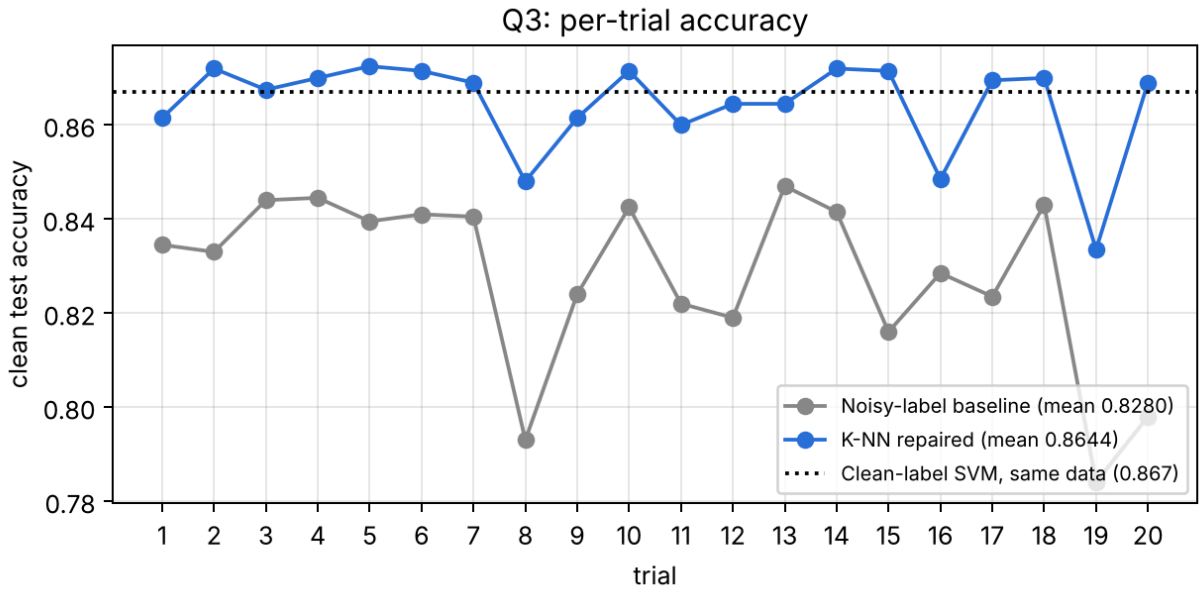
k	**2**	**3**	**4**	**6**	**10**
SSE	15,222	6,826	2,794	2,397	1,915



With k < 4, whole blobs get merged: k = 2 groups three blobs together, and k = 3 merges the two closest ones. At k = 4 the clusters match the four true blobs (~1,000 points each). With k > 4, real blobs get chopped into pieces, and which ones get split depends on the initialization. SSE keeps going down as k grows, but the big drop stops at k = 4 (the elbow), which matches the four generating groups.

Q3: Noisy labels and K-NN repair

Train and test sets are generated once (seed 0) and kept fixed; test labels stay clean. Each trial flips +1 labels with p = 0.35 and −1 labels with p = 0.20 (one draw per point, based on the original label). For repair, each point gets the majority label of its 5 nearest other training points, all votes taken from the same noisy labels. Distance ties are broken by lower index. Bothmodels are RBF SVMs (C = 1, γ = 0.01) trained on the same noisy labels in each trial.



Mean test accuracy: **baseline 0.8280**, **K-NN 0.8644**, difference **+0.0364**. K-NN was better in all 20 trials (per-trial numbers are in noise_comparison.csv).
Repair helped because the noise is random, so a flipped point usually sits among neighbors with the correct label, and the vote fixes it. That gives the SVM smoother, more consistent labels, and it gets close to the 0.867 from training on clean labels. It could hurt when there are real minority points near the class boundary, which get overwritten. It would also fail if noise were clustered or close to 50%: here positives flip more often (35%), so overlap regions lean negative after repair.